PHYS 230 - Electricity and Magnetism
Lecture 28

Instructor: Saeed Rastgoo

UNIVERSITY OF ALBERTA

Outline

7. Magnetic Field and Magnetic Forces
   - Magnetic Field Lines and Magnetic Flux (Continued)
   - Motion of Charged Particles in a Magnetic Field
   - Magnetic Force on a Current-carrying Conductor

All figures in these slides are from one of the following sources, unless cited otherwise:

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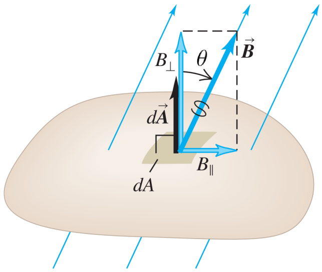
Magnetic Flux

Infinitesimal magnetic flux $d\phi_B$ of a magnetic field **B** through an infinitesimal area dA is

$$\boxed{d\phi_B = \mathbf{B} \cdot d\mathbf{A}} = B\, dA \cos\left(\theta_{\mathbf{B},\, d\mathbf{A}}\right)$$

Figure **27.15** The magnetic flux through an area element dA is defined to be $d\Phi_B = B_\perp dA$.



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If the area belongs to a closed surface, then $d\mathbf{A}$ is in the direction of the unit vector $\mathbf{n}$ pointing outwards; for open surfaces you choose the direction

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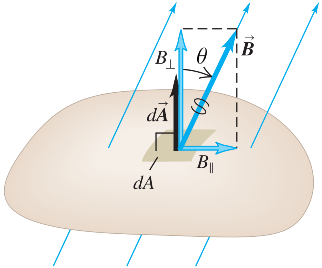

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The total magnetic flux is then

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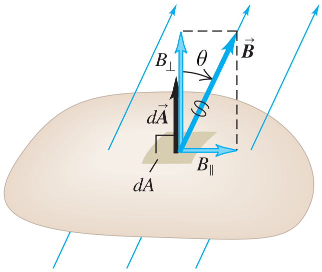

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SI units of ϕ_B is **weber**, Wb

$$[\phi_B] = [\mathbf{B}]\,[\mathbf{A}] = \mathrm{T} \cdot \mathrm{m}^2 = \mathrm{Wb}$$

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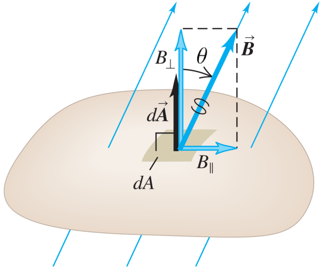

Gauss's Law For Magnetism

Because there is no single magnetic charge (monopole) in nature, Gauss's law for magnetic becomes

$$\phi_B = \boxed{\oint \mathbf{B} \cdot d\mathbf{A} = 0}$$

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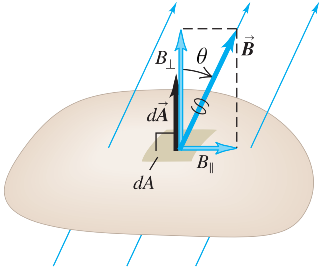
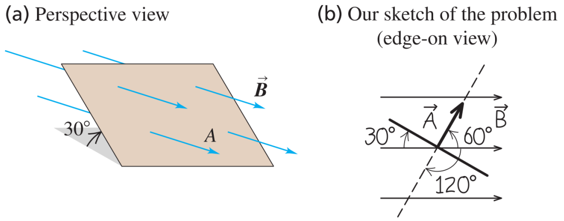


Exercise 1 https://epoll.srv.ualberta.ca – Code: MBD

Figure shows a perspective view of a flat surface with area $3.0\,\text{cm}^2$ in a uniform magnetic field **B**. The magnetic flux through this surface is $+0.90$ mWb. Find the magnitude of the magnetic field and the direction of the area vector **A**.



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Charged Particles in Magnetic Field

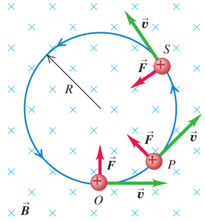
According to

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$$

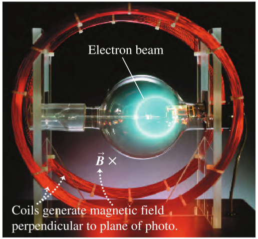
the force **F** on a charged particle q in a magnetic field **B** is perpendicular to the particle's velocity **v**

(a) The orbit of a charged particle in a uniform magnetic field

A charge moving at right angles to a uniform $\vec{B}$ field moves in a circle at constant speed because $\vec{F}$ and $\vec{v}$ are always perpendicular to each other.



(b) An electron beam (seen as a white arc) curving in a magnetic field



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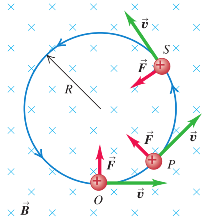

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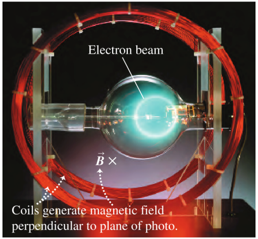

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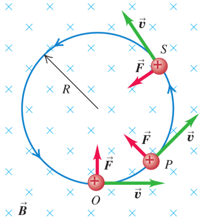

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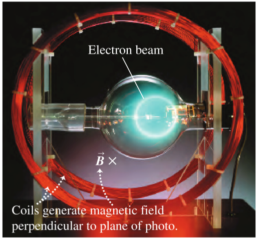

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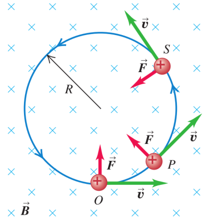

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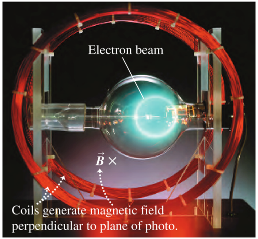

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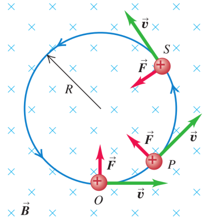

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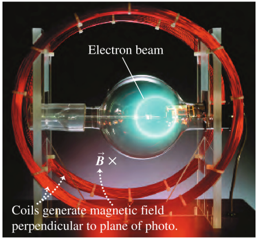

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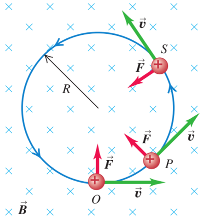

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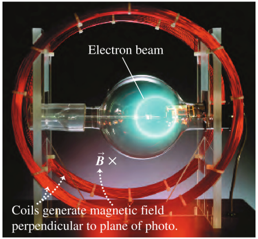

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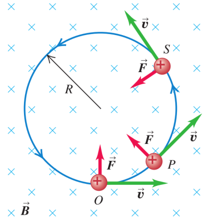

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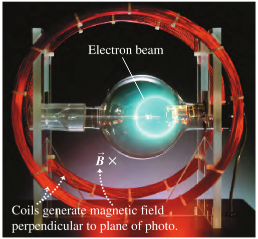

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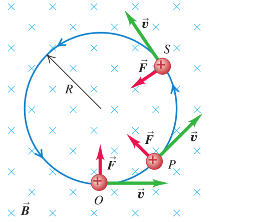

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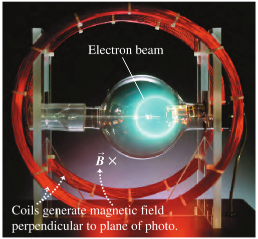

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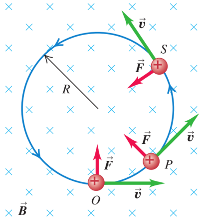

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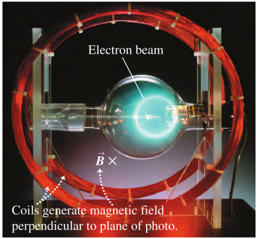

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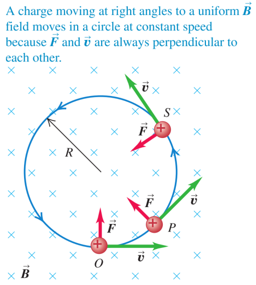

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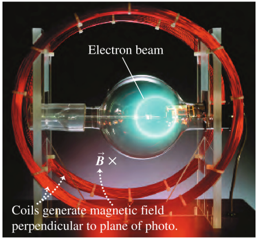

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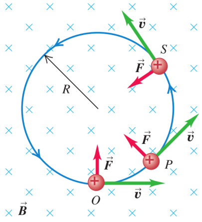

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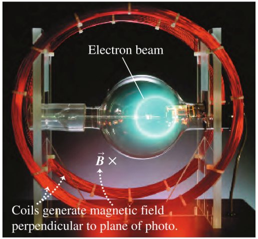

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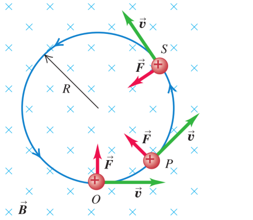

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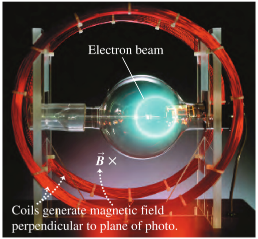

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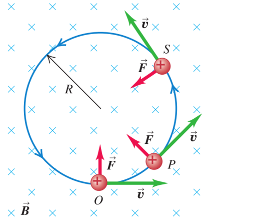

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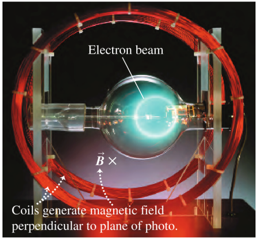

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According to Newton's laws

$$|\mathbf{F}| = \frac{mv_\perp^2}{r} \Rightarrow |q|\, v_\perp B \sin\left(\underbrace{\theta_{\mathbf{v},\,\mathbf{B}}}_{90^\circ}\right) = m\frac{v_\perp^2}{r}$$

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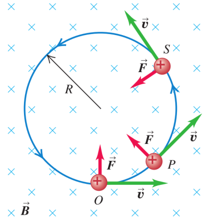

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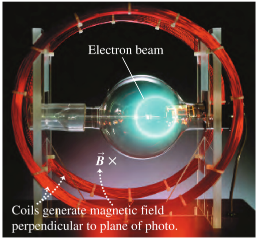

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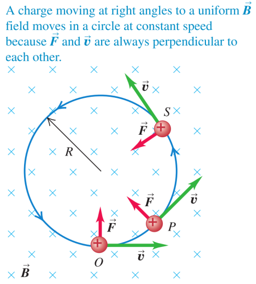
(a) The orbit of a charged particle in a uniform magnetic field



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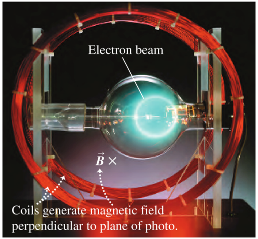

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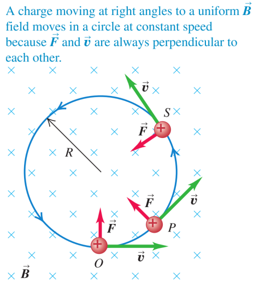

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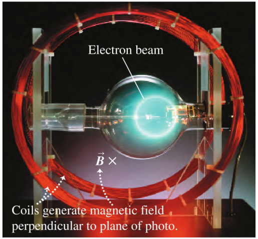

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$$v_\perp = r\omega$$

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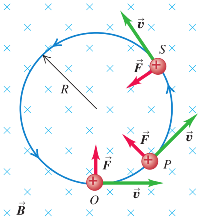

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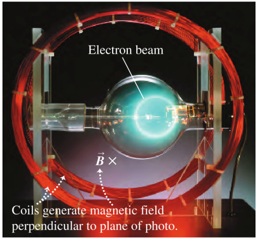

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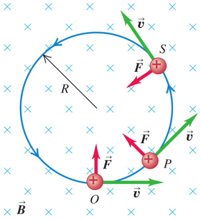

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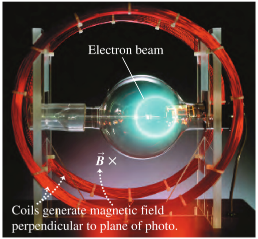

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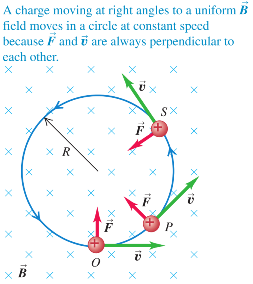

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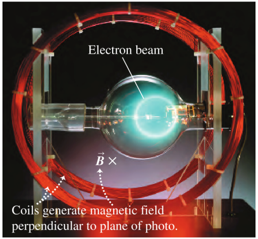

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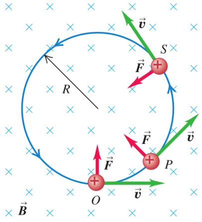

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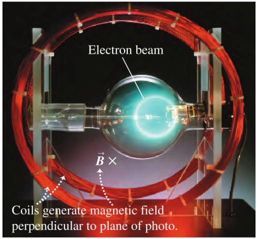

Charged Particles in Magnetic Field

The angular frequency f is called the cyclotron frequency and is independent of the radius of rotation r

$$\omega = 2\pi f$$

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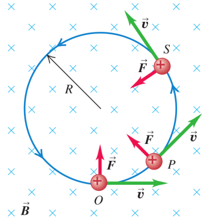

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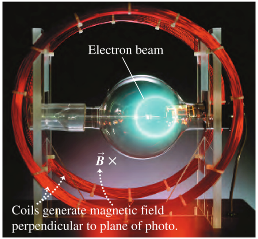

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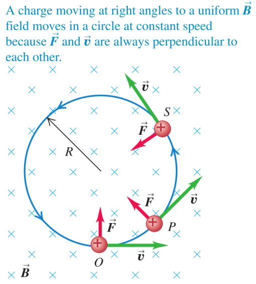
The angular frequency *f* is called the cyclotron frequency and is independent of the radius of rotation *r*

$$\omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi}$$

(a) The orbit of a charged particle in a uniform magnetic field



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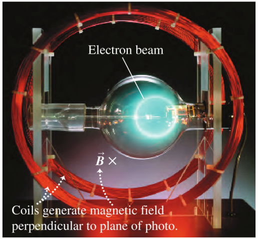

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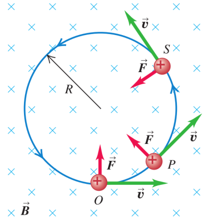

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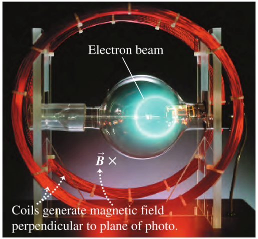
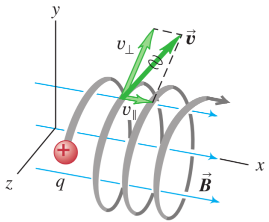


Charged Particles in Magnetic Field

There is no force in the direction of $\mathbf{v}_{\parallel}$

$$\omega = 2\pi f$$

This particle's motion has components both parallel ($v_{\parallel}$) and perpendicular ($v_{\perp}$) to the magnetic field, so it moves in a helical path.



Charged Particles in Magnetic Field

There is no force in the direction of $\mathbf{v}_{\|}$

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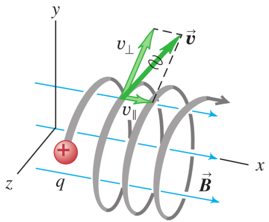
This particle's motion has components both parallel ($v_{\|}$) and perpendicular ($v_{\perp}$) to the magnetic field, so it moves in a helical path.



Charged Particles in Magnetic Field

There is no force in the direction of $\mathbf{v}_{\parallel}$

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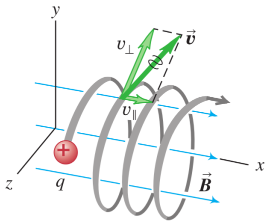

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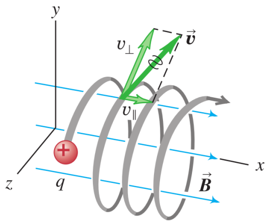


Charged Particles in Magnetic Field

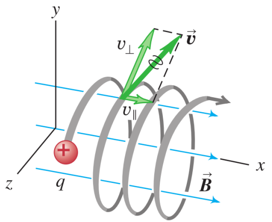
There is no force in the direction of $\mathbf{v}_{||}$

$$\omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi} \Rightarrow \boxed{f = \frac{1}{2\pi}\frac{|q|}{m}B}$$

So if $\mathbf{v}_{||} \neq 0$, then $\mathbf{v}_{||}$ will remains constant, and creates a motion perpendicular to the plane of circular motion

$\Rightarrow$ the particle moves in a helix

This particle's motion has components both parallel ($v_{\|}$) and perpendicular ($v_{\perp}$) to the magnetic field, so it moves in a helical path.



Exercise 2

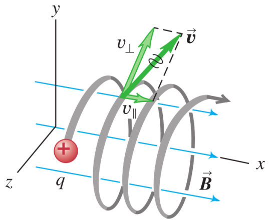
https://epoll.srv.ualberta.ca – Code: MBD

A magnetron in a microwave oven emits electromagnetic waves with frequency $f = 2450$ MHz. What magnetic field strength is required for electrons to move in circular paths with this frequency?

Exercise 3 https://epoll.srv.ualberta.ca – Code: MBD

In a situation like that shown in figure, the charged particle is a proton ($q = 1.6 \times 10^{-19}$ C, $m = 1.67 \times 10^{-27}$ kg) and the uniform, 0.5 T magnetic field is directed along the x-axis. At $t = 0$ the proton has velocity components $v_x = 1.5 \times 10^5$ m/s , $v_y = 0$, and $v_z = 2 \times 10^5$ m/s. Only the magnetic force acts on the proton. (a) At $t = 0$, find the force (vector) on the proton and its acceleration (vector). (b) Find the radius of the resulting helical path, the angular speed of the proton, and the pitch of the helix (the distance traveled along the helix axis per revolution).

This particle's motion has components both parallel ($v_\parallel$) and perpendicular ($v_\perp$) to the magnetic field, so it moves in a helical path.



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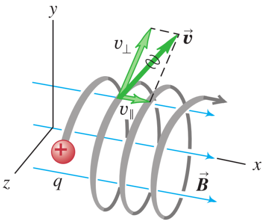

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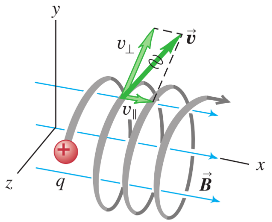


Outline

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

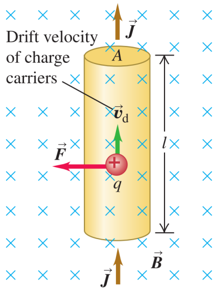
Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

Force on a Positive Charge in a Wire

A conducting wire, with length ℓ, cross-sectional area A; the current is from bottom to the top; **B** perpendicular to the plane of the diagram and into the plane

Figure 27.25 Forces on a moving positive charge in a current-carrying conductor.



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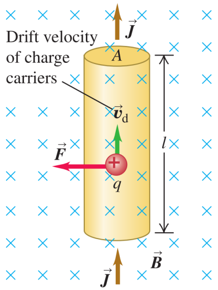
With drift velocity $\mathbf{v}_d$ is upward, perpendicular to $\mathbf{B}$, the force on a charge q is

$$\mathbf{F} = q\mathbf{v}_d \times \mathbf{B} = q\frac{\boldsymbol{\ell}}{t} \times \mathbf{B} = \frac{q}{t}\boldsymbol{\ell} \times \mathbf{B}$$

Figure 27.25 Forces on a moving positive charge in a current-carrying conductor.



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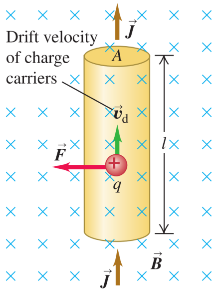

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- t is the time it takes for q to go from the bottom to the top of wire with length ℓ

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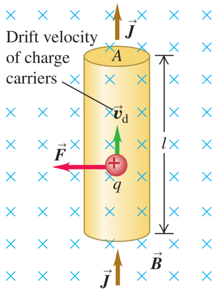

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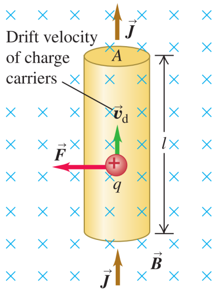
But since $\frac{q}{t} = I$ is the current in the wire we get

$$\boxed{\mathbf{F} = I\boldsymbol{\ell} \times \mathbf{B}}$$

Figure **27.25** Forces on a moving positive charge in a current-carrying conductor.



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- t is the time it takes for q to go from the bottom to the top of wire with length ℓ

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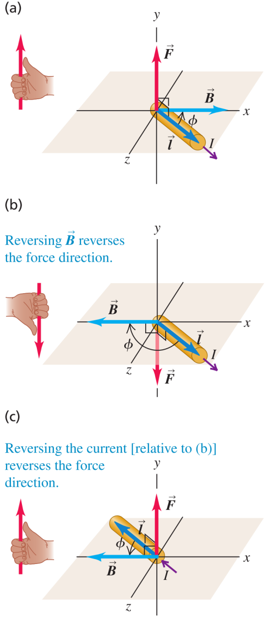
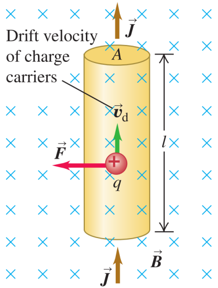


Force on a Positive Charge in a Wire

For the infinitesimal force on an infinitesimal length $d\ell$ of a wire we have

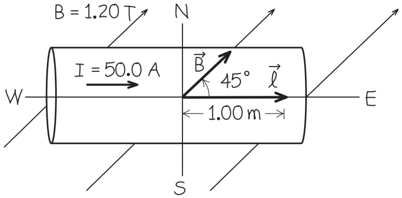
$$\boxed{d\mathbf{F} = I\, d\boldsymbol{\ell} \times \mathbf{B}}$$

Figure 27.25 Forces on a moving positive charge in a current-carrying conductor.

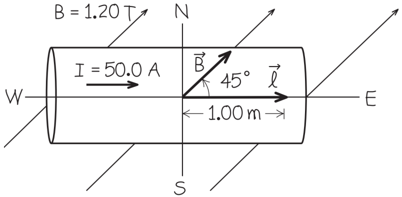


Exercise 4 https://epoll.srv.ualberta.ca – Code: MBD

A straight horizontal copper rod carries a current of 50.0 A from west to east in a region between the poles of a large electromagnet. In this region there is a horizontal magnetic field toward the northeast (that is, $45°$ north of east) with magnitude 1.20 T. (a) Find the vector force on a 1.00 m section of rod. (b) While keeping the rod horizontal, how should it be oriented to maximize the magnitude of the force? What is the force vector in this case?

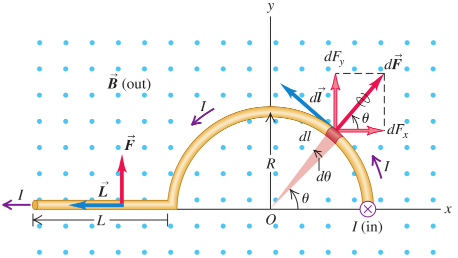


Exercise 4 https://epoll.srv.ualberta.ca – Code: MBD

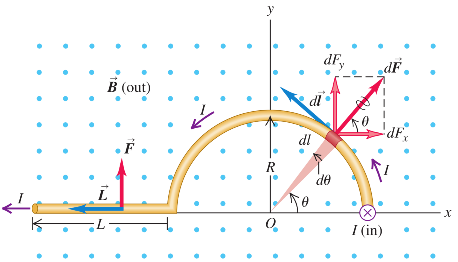


Exercise 5 https://epoll.srv.ualberta.ca – Code: MBD

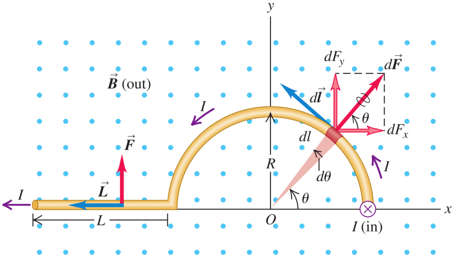
In this figure, the magnetic field **B** is uniform and perpendicular to the plane of the figure, pointing out of the page. The conductor, carrying current *I* to the left, has three segments: (1) a straight segment with length *L* perpendicular to the plane of the figure, (2) a semicircle with radius *R*, and (3) another straight segment with length *L* parallel to the *x*-axis. Find the total magnetic force on this conductor.



Exercise 5 https://epoll.srv.ualberta.ca – Code: MBD



Exercise 5 https://epoll.srv.ualberta.ca – Code: MBD



Exercise 5 https://epoll.srv.ualberta.ca – Code: MBD